

Mark Scheme — Practice Paper 7 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 Term = $\binom{7}{4}(3)^3(-2x)^4$ (M1 A1) = $35 \times 27 \times 16 = 15120$ (A1). [3]

2 $\sin x = \pm \frac{1}{2}$ (M1 A1, both signs). $x = \frac{\pi}{6}, \frac{5\pi}{6}$ (A1), $\frac{7\pi}{6}, \frac{11\pi}{6}$ (A1). [4]

3 $ar = 20, ar^4 = 2.5 \Rightarrow r^3 = \frac{1}{8}$ (M1) $\Rightarrow r = \frac{1}{2}$ (A1), $a = 40$ (A1). $S_\infty = \frac{40}{1 - \frac{1}{2}} = 80$ (A1). [4]

4(a) $3(x^2 + 4x) + 5 = 3[(x + 2)^2 - 4] + 5$ (M1) = $3(x + 2)^2 - 7$ (A1 $b = 2$, A1 $c = -7$). [3]

4(b) $(x + 2)^2 = \frac{7}{3} \Rightarrow x = -2 \pm \sqrt{\frac{7}{3}}$ (M1) = $-2 \pm \frac{\sqrt{21}}{3}$ (A1; either form). [2]

5(a) $2r + r\theta = 40 \Rightarrow \theta = \frac{40 - 2r}{r}$ (M1). $A = \frac{1}{2}r^2\theta = \frac{1}{2}r(40 - 2r) = 20r - r^2$ (A1 AG). [2]

5(b) $\frac{dA}{dr} = 20 - 2r = 0 \Rightarrow r = 10$ (M1 A1). $\theta = \frac{40 - 20}{10} = 2$ (A1). [3]

5(c) $\frac{d^2A}{dr^2} = -2 < 0$, maximum; $A = 100$ (B1, both). [1]

6(a) Gradient of l is $\frac{1}{2}$ (B1), so perpendicular gradient = -2 (M1). $y + 1 = -2(x - 3) \Rightarrow y = -2x + 5$ (A1). [3]

6(b) Solve $y = -2x + 5$ with $y = \frac{1}{2}x + 2$ (M1): $-2x + 5 = \frac{1}{2}x + 2 \Rightarrow x = \frac{6}{5}$ (A1). Foot = $(1.2, 2.6)$ (A1). [3]

7(a) $(x - 2)^2 + 3$ (B1 B1). [2]

7(b) $f(x) \geq 3$ (B1). [1]

7(c) $x - 2 = \pm\sqrt{y - 3}$; since $x \leq 2$ take negative root (M1): $f^{-1}(x) = 2 - \sqrt{x - 3}$ (A1). Domain $x \geq 3$ (B1). [3]

7(d) $(x - 2)^2 = 9 \Rightarrow x = -1$ (only; $x = 5$ is outside the domain) (B1). [1]

8(a) $\frac{dy}{dx} = 3x^{1/2} - 9$ (B1). $= 0 \Rightarrow \sqrt{x} = 3 \Rightarrow x = 9$ (M1); $y = 2(27) - 81 + 1 = -26$: $(9, -26)$

(A1). $\frac{d^2y}{dx^2} = \frac{3}{2}x^{-1/2} > 0$, minimum (A1). [4]

8(b) At $x = 4$: $y = 16 - 36 + 1 = -19$, $\frac{dy}{dx} = 6 - 9 = -3$ (B1 B1). $y + 19 = -3(x - 4) \Rightarrow y = -3x - 7$ (A1). [3]

9(a) $\frac{(1 + \sin \theta)^2 + \cos^2 \theta}{\cos \theta(1 + \sin \theta)}$ (M1) = $\frac{1 + 2 \sin \theta + \sin^2 \theta + \cos^2 \theta}{\cos \theta(1 + \sin \theta)} = \frac{2 + 2 \sin \theta}{\cos \theta(1 + \sin \theta)}$ (M1) = $\frac{2}{\cos \theta}$ (A1 AG). [3]

9(b) $\cos \theta = \frac{2}{5}$ (M1). $\theta = 66.4^\circ$ (A1), 293.6° (A1). [3]

9(c) $2x$ ranges over 0° to 720° — two full cycles, each giving 2 solutions (M1): 4 solutions (A1). [2]

10(a) $\int 3(2x + 1)^{-1/2} dx = \frac{3(2x + 1)^{1/2}}{\frac{1}{2} \times 2} = 3\sqrt{2x + 1}$ (M1 A1). $[3\sqrt{2x + 1}]_0^4 = 9 - 3 = 6$ (A1). [3]

10(b) $V = \pi \int_0^5 (3x + 1) dx$ (M1) = $\pi [\frac{3}{2}x^2 + x]_0^5$ (A1) = $\pi(37.5 + 5) = 42.5\pi$ (A1; = $\frac{85}{2}\pi$). [3]

10(c) $[2\sqrt{x} - x]_1^4 = (4 - 4) - (2 - 1)$ (M1) = -1 (A1). [2]

11(a)(i) $S_{10} = 5(2a + 9d) = 165 \Rightarrow 2a + 9d = 33$ (B1). $S_{20} = 165 + 465 = 630 \Rightarrow 10(2a + 19d) = 630 \Rightarrow 2a + 19d = 63$ (M1 A1). Subtract: $10d = 30 \Rightarrow d = 3, a = 3$ (A1). **[4]**

11(a)(ii) $u_{25} = 3 + 24 \times 3 = 75$ (B1). **[1]**

11(b) $a(1 + r + r^2) = 21$ and $\frac{a}{1-r} = 24$ (B1). $24(1-r)(1+r+r^2) = 21 \Rightarrow 24(1-r^3) = 21$ (M1 A1) $\Rightarrow r^3 = \frac{1}{8} \Rightarrow r = \frac{1}{2}, a = 12$ (A1). **[4]**

12(a) $(x-1)^2 + (y-3)^2 = 5$: centre $(1, 3)$ (B1), radius $\sqrt{5}$ (B1). **[2]**

12(b) $9 + 16 - 6 - 24 + 5 = 0$ ✓ (B1). Radius gradient $\frac{4-3}{3-1} = \frac{1}{2}$, tangent gradient -2 (M1). $y - 4 = -2(x - 3) \Rightarrow y = -2x + 10$ (A1). **[3]**

12(c) $A(5, 0)$ (B1), $B(0, 10)$ (B1). Area $= \frac{1}{2} \times 5 \times 10 = 25$ (B1). **[3]**

Total: 75